



1.2 Sequences and series

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

An arithmetic sequence has constant common difference d : $u_n = u_1 + (n-1)d$, with sum $S_n = n/2(2u_1 + (n-1)d)$. A geometric sequence has constant common ratio r : $u_n = u_1 r^{n-1}$, with sum $S_n = u_1(r^n - 1)/(r - 1)$ for $r \neq 1$. If $|r| < 1$, the infinite geometric series converges to $S_\infty = u_1/(1-r)$. These models describe contexts with constant additive or multiplicative change, such as seating arrangements or bouncing balls.

Worked example: follow the reasoning

A theatre has 20 rows; row 1 has 15 seats, and each subsequent row has 3 more seats than the previous row. Find the number of seats in row 20 and the total number of seats.
Solution: $u_1 = 15$, $d = 3$, $n = 20$. $u_{20} = 15 + 19 \times 3 = 72$ seats. $S_{20} = 20/2 \times (2 \times 15 + 19 \times 3) = 10 \times 87 = 870$ seats.

Quick check · multiple choice

A geometric sequence begins 8, 12, 18, 27, ... What is the common ratio?

- A 1.5
- B 4
- C 0.67
- D 2

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A ball is dropped from a height of 4 m. After each bounce it rises to 60% of the height it fell from. (a) Find the height reached after the 4th bounce. (b) Find the total vertical distance the ball travels if it continues bouncing indefinitely. [6]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $h_n = 4 \times 0.6^n$, so $h_4 = 4 \times 0.6^4 = 4 \times 0.1296 = 0.5184 \text{ m} \approx 0.518 \text{ m}$. (b) Total distance = initial drop + $2 \times$ (sum of all bounce heights, since each bounce is travelled up and down) = $4 + 2 \times [4 \times 0.6 / (1 - 0.6)] = 4 + 2 \times 6 = 16 \text{ m}$.

Quick check answer

A. $r = 12/8 = 1.5$ (and $18/12 = 1.5$, confirming a constant ratio).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.